MTD-HealthNet: A Moving Target Defense Model for Securing Hospital Networks

PROJECT REPORT

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BONAFIDE CERTIFICATE

This is to certify that this project entitled **"MTD-HealthNet: A Moving Target Defense Model for Securing Hospital Networks"** submitted by **Burla Rushyendra Reddy (CB.EN.U4CYS22015), Karri Jeeshitha (CB.EN.U4CYS22032), Vajjula Satya Siddardha (CB.EN.U4CYS22063), Yallanuru Kishan Sai (CB.EN.U4CYS22064)** in partial fulfillment of the requirements for the award of the **Degree of Bachelor of Technology** in **Computer Science And Engineering (Cyber Security)** is a bonafide record of the work carried out by them under my guidance and supervision at Amrita School of Computing, Coimbatore.

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DECLARATION

We hereby declare that this project entitled **"MTD-HealthNet: A Moving Target Defense Model for Securing Hospital Networks"** is a record of the original work done by us under the guidance of **Mr. Akhil K J**, Assistant Professor, TIFAC-CORE in Cyber Security, Amrita Vishwa Vidyapeetham and that this work has not formed the basis for the award of any degree, diploma, associateship, fellowship, or any other similar title, to any candidate in any university, to the best of our knowledge.

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Abstract

This project addresses a critical challenge in modern **Network Security** by focusing on how hospital networks can be better protected against evolving cyber threats. With the rise in ransomware incidents, unauthorized access attempts, and data breaches targeting healthcare institutions, it has become essential to move beyond traditional, static security measures. This work explores the use of **Moving Target Defense (MTD)** to make hospital LANs more dynamic, adaptive, and resistant to reconnaissance-based attacks.

The motivation behind this study stems from the increasing dependency of hospitals on uninterrupted digital communication and the need to secure sensitive patient information. Static IP configurations and predictable network behaviour make hospital networks easy targets for attackers. To address these issues, this project examines key MTD techniques such as **IP hopping**, **port hopping**, and **NAT randomization**, evaluating how they can enhance security without disrupting essential medical operations.

Based on the gaps identified in existing literature—such as maintaining session continuity, supporting legacy systems, and avoiding service interruption—this project proposes **MTD-HealthNet**, a dynamic defense model tailored for hospital environments. The system introduces controlled randomness into network addressing while ensuring that core services like DNS resolution, device communication, and data flow remain stable.

The prototype was implemented and tested within a simulated hospital environment using **Mininet**, where multiple zones and devices were configured to reflect real-world conditions. Security tools including **Wireshark** and **Nmap** were used to validate network stability, measure performance, and analyze security improvements. The results demonstrated significant reduction in the attack surface, improved resilience, and almost zero packet loss during reconfiguration cycles.

Overall, **MTD-HealthNet** shows that dynamic and adaptive defense mechanisms can strengthen hospital networks without requiring major hardware upgrades. The solution is practical, scalable, compliant with regulations, and suitable for modern healthcare infrastructures. It also opens promising directions for future work in adaptive cyber defense, SDN-based security, and IoT protection in healthcare systems.

Keywords: Network Security, Moving Target Defense, Hospital LAN, IP Hopping, Port Hopping, NAT Randomization, Mininet, Dynamic Defense, DNS Resolution, SDN, Healthcare Cybersecurity, Wireshark, Nmap, HIPAA, GDPR.

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List of Abbreviations

LAN	–	Local Area Network
MTD	–	Moving Target Defense
IP	–	Internet Protocol
SDN	–	Software-Defined Networking
NAT	–	Network Address Translation
DNAT	–	Destination Network Address Translation
SNAT	–	Source Network Address Translation
MAC	–	Media Access Control
TCP	–	Transmission Control Protocol
UDP	–	User Datagram Protocol
TTL	–	Time To Live
DNS	–	Domain Name System
DHCP	–	Dynamic Host Configuration Protocol
HTTP	–	Hypertext Transfer Protocol
REST	–	Representational State Transfer
API	–	Application Programming Interface
EHR	–	Electronic Health Record
HL7	–	Health Level Seven (Healthcare Messaging Standard)
DICOM	–	Digital Imaging and Communications in Medicine
OVS	–	Open vSwitch
OF	–	OpenFlow Protocol
RTT	–	Round Trip Time
PDR	–	Packet Delivery Ratio
IDS	–	Intrusion Detection System
IPS	–	Intrusion Prevention System
Wireshark	–	Network Protocol Analyzer
Nmap	–	Network Mapper (Port Scanning Tool)
HIPAA	–	Health Insurance Portability and Accountability Act
GDPR	–	General Data Protection Regulation
VM	–	Virtual Machine

Chapter 1

Introduction

1.1 Motivation

Modern healthcare infrastructures depend heavily on interconnected digital systems such as Electronic Health Records (EHR), patient monitoring devices, imaging equipment, and hospital communication platforms. While these technologies significantly improve healthcare efficiency and patient care, they also introduce new cybersecurity risks. Hospital networks have increasingly become attractive targets for attackers because they contain sensitive patient data and support critical medical services.

Many healthcare networks still operate with relatively static configurations where IP addresses, ports, and routing paths remain unchanged for long periods. Such predictability allows attackers to perform reconnaissance, map the network structure, and reuse this information to launch targeted attacks such as ransomware, unauthorized data access, or service disruption.

The motivation for this project is to design a security framework that makes hospital networks more resilient against such attacks without disrupting normal clinical operations. Instead of relying only on traditional reactive defenses, our approach focuses on dynamically changing the network's attack surface. By continuously modifying key network parameters, the system aims to reduce predictability and protect critical healthcare services while ensuring stable communication for medical staff and devices.

1.2 Problem Statement

A major security limitation in many hospital Local Area Networks (LANs) is their static network configuration. When IP addresses, port assignments, and routing structures remain constant, attackers can conduct reconnaissance once and repeatedly exploit the

same information. This exposes hospital networks to serious risks including data breaches, ransomware attacks, and interruptions to life-critical medical systems.

This project addresses this issue by designing and implementing a Moving Target Defense (MTD) framework named **MTD-HealthNet**. The proposed system dynamically modifies network parameters such as IP addresses and Network Address Translation (NAT) rules. By periodically changing these parameters, previously collected reconnaissance information becomes invalid, thereby reducing the effectiveness of targeted attacks while preserving normal network functionality for hospital systems.

1.3 Threat Model

In the proposed threat model, we consider an adversary who performs reconnaissance before attempting exploitation. The attacker may conduct activities such as network scanning, device fingerprinting, IP spoofing, lateral movement, and malicious payload delivery (for example ransomware deployment). In traditional static networks, such activities are easier because the network environment remains predictable and unchanged.

The proposed **MTD-HealthNet** system mitigates these risks by dynamically altering network configurations. Techniques such as IP hopping and dynamic Network Address Translation using DNAT and SNAT are employed to continuously modify the network environment. As a result, information gathered during reconnaissance quickly becomes outdated, significantly limiting the attacker's ability to exploit the system.

In addition to these dynamic network mechanisms, the system also applies policy enforcement and lightweight cryptographic verification to maintain secure communication between network entities.

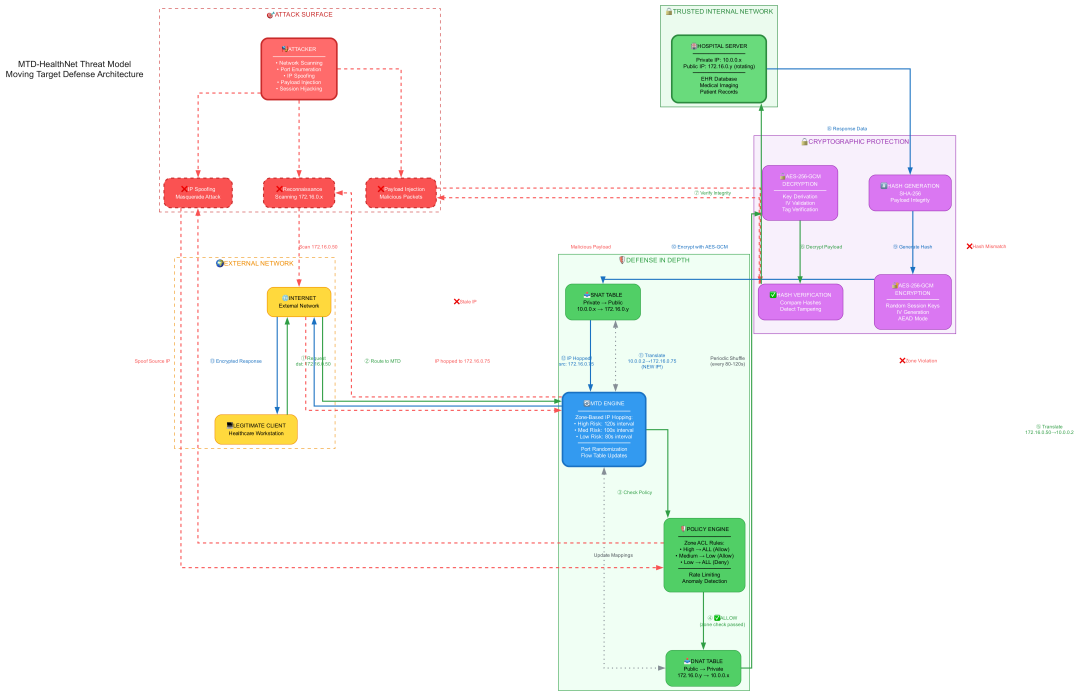

Figure 1.1: **Threat Model and Defense Architecture of MTD-HealthNet.** The diagram illustrates the attacker reconnaissance process and the security mechanisms implemented in the proposed system, including Moving Target Defense through IP hopping, dynamic NAT translation (DNAT and SNAT), policy enforcement, and secure communication verification.

1.4 Goals

The primary objectives of this project are summarized below:

- Design and implement a Moving Target Defense framework specifically tailored for hospital networks, referred to as **MTD-HealthNet**.
- Introduce dynamic security mechanisms such as IP hopping and NAT table randomization in order to reduce the predictability of the network attack surface.
- Ensure uninterrupted communication for essential medical systems and healthcare services even when network parameters change dynamically.
- Evaluate the effectiveness of the proposed system through simulated attack scenarios and measure both security improvements and network performance.

1.5 Research Methodology

To evaluate the effectiveness of the proposed system, the study is guided by the following research questions:

1. What are the common security weaknesses present in traditional hospital LAN architectures?
2. How effectively can Moving Target Defense techniques reduce attacker reconnaissance and exploitation attempts?
3. What is the impact of dynamic network changes on performance metrics such as latency, packet loss, and routing stability?
4. How can active network sessions remain stable even when IP addresses change dynamically?

To address these research questions, the following methodological steps were followed:

- **Analysis of hospital network architectures:** Typical hospital LAN environments were studied to identify security weaknesses such as static IP addressing, predictable routing paths, and limited network obfuscation.
- **Implementation of Moving Target Defense mechanisms:** A simulated hospital network environment was created using Mininet. Within this environment, mechanisms such as IP hopping and dynamic NAT translation were implemented to observe how these techniques affect attacker reconnaissance.
- **Security and performance evaluation:** Network monitoring tools such as Wireshark, ping tests, and Nmap scans were used to analyze system performance before and after applying the MTD mechanisms.
- **Session continuity validation:** Continuous communication tests were conducted to verify that active network sessions remain stable even when IP addresses change dynamically.

1.6 Structure of the Project

The remainder of this report is organized into seven chapters.

Chapter 2 presents the background concepts related to hospital network security and Moving Target Defense.

Chapter 3 provides a detailed literature survey discussing existing research and identifying gaps addressed by this work.

Chapter 4 explains the design and architecture of the proposed **MTD-HealthNet** system.

Chapter 5 describes the main contributions of the project and highlights the innovations introduced in the proposed approach.

Chapter 6 presents the experimental results and performance analysis obtained from the simulated network environment.

Chapter 7 concludes the study and summarizes the key findings.

Finally, Chapter 8 discusses potential future work and possible improvements to the proposed system.

Chapter 2

Background

This chapter introduces the fundamental networking concepts and technologies that form the foundation of the proposed system. To understand the motivation behind the design of MTD-HealthNet, it is important to examine how traditional hospital networks operate and why their static configurations make them vulnerable to cyber threats. This chapter also introduces the concept of Moving Target Defense (MTD) as a proactive security approach. Furthermore, it discusses common vulnerabilities in healthcare LAN environments and presents the tools and methodologies used to design and evaluate the proposed system.

2.1 Overview of Hospital Network Security

Modern healthcare environments rely heavily on interconnected digital systems such as Electronic Health Record (EHR) databases, patient monitoring devices, imaging systems, and administrative computers. These devices communicate through a shared Local Area Network (LAN), which is typically segmented to separate critical medical equipment from general administrative systems.

Despite these segmentation practices, many hospital networks still operate using static configurations where IP addresses, routing structures, and network parameters rarely change. Static configurations simplify network management and ensure compatibility with legacy medical equipment. However, this stability also creates a predictable environment that attackers can exploit.

Because the network topology remains unchanged for long periods, attackers can perform reconnaissance once and retain a reliable blueprint of the system. This predictability significantly increases the risk of cyber incidents such as data breaches, ransomware attacks, and unauthorized access to sensitive patient information.

2.2 Static LAN Configurations: The Core Problem

Traditional hospital LANs commonly rely on fixed IP assignments, predictable routing paths, and static access policies. From a network management perspective, these configurations simplify troubleshooting and system maintenance. However, they introduce serious security concerns.

Several key weaknesses arise from static network architectures:

- Fixed IP addressing enables attackers to repeatedly target the same critical systems once they have been identified.
- Predictable network structures provide consistent entry points that attackers can exploit.
- The absence of dynamic reconfiguration allows reconnaissance data collected by attackers to remain valid for extended periods of time.

As a result, attackers can conduct extensive scanning activities, identify valuable targets, and plan attacks without worrying about changes in the network environment.

2.3 Moving Target Defense (MTD) as a Solution

To address the limitations of static network architectures, this project explores the concept of Moving Target Defense (MTD). Unlike traditional security mechanisms such as firewalls or antivirus software, which act as static defensive barriers, MTD actively modifies the attack surface of a system.

By dynamically changing network parameters, MTD makes it difficult for attackers to maintain an accurate understanding of the network structure. As a result, reconnaissance information becomes obsolete, forcing attackers to repeatedly restart their scanning activities.

In the proposed hospital network framework, referred to as **MTD-HealthNet**, MTD techniques are implemented through the following mechanisms:

- Periodic IP hopping, which changes host IP addresses at predefined intervals.
- Dynamic Network Address Translation (NAT) randomization, which hides the internal private network structure behind rotating public IP addresses.

These mechanisms significantly reduce the effectiveness of attacker reconnaissance and increase the effort required to successfully exploit the network.

2.4 Hospital Network Vulnerabilities and Attack Vectors

Hospital networks often contain legacy medical devices that cannot easily be updated or patched. These devices frequently operate within static network environments, making them susceptible to several common attack vectors.

The following attack techniques are particularly relevant in healthcare environments:

- **Network Reconnaissance:** Attackers use scanning tools to discover active hosts, identify open ports, and detect vulnerable services.
- **Lateral Movement:** After compromising an initial device, attackers move through the network to reach sensitive systems such as EHR servers.
- **IP Spoofing:** Attackers forge source IP addresses in order to bypass basic access control mechanisms and impersonate trusted devices.
- **Ransomware Deployment:** Once attackers understand the network layout, ransomware can be deployed directly against critical hospital infrastructure.

These attack strategies are particularly effective in static networks where system configurations remain unchanged.

2.5 Methodology and Evaluation Tools

To evaluate the performance and security effectiveness of the proposed MTD-HealthNet system, a combination of network simulation tools, custom scripts, and monitoring techniques were used. These tools enabled the testing of host-to-host communication, dynamic IP hopping, NAT behavior, and controller-driven security policies within a Software Defined Networking (SDN) environment.

2.5.1 Mininet Network Emulator

Mininet was used as the primary network emulation platform to simulate the hospital network environment. The simulated topology consisted of multiple hosts representing medical devices and administrative systems, an OpenFlow switch, and a centralized SDN controller.

Mininet allowed the creation of a realistic virtual network environment in which routing behavior, link characteristics, and host communication could be controlled programmatically.

The following evaluations were performed using Mininet:

- Host-to-host connectivity testing during IP rotations.
- Verification of packet forwarding behavior under NAT translation.
- Evaluation of communication stability while IP hopping was active.
- Secure data transmission testing between network nodes.

Standard connectivity tests such as `pingall` were initially used to establish baseline connectivity. However, due to the dynamic NAT architecture and IP hopping mechanisms, ICMP-based tests did not always reflect the success of application-level communication.

2.5.2 Ryu SDN Controller Monitoring

The proposed MTD controller was implemented using the Ryu SDN framework. The Ryu controller was responsible for dynamically managing network behavior through OpenFlow.

Its primary responsibilities included:

- Installing OpenFlow flow rules to control packet forwarding.
- Managing NAT mappings between private IP addresses (10.0.0.0/24) and dynamically assigned public MTD IP addresses (172.16.0.0/16).
- Enforcing security policies through zone-based access control mechanisms.
- Triggering periodic IP hopping operations based on predefined rotation intervals.

Controller logs and REST API responses were analyzed to monitor policy decisions, routing changes, and potential system exceptions during execution.

2.5.3 REST API-Based Testing and cURL Validation

To evaluate real data transmission between hosts, the system exposed a `/sim/secure_transfer` API endpoint. Repeated `cURL` requests were used to initiate secure communication between simulated hosts.

These tests verified:

- Source and destination host validation according to policy rules.
- Successful transmission of encrypted payloads.
- Proper delivery confirmation through controller logs.
- Correct HTTP response acknowledgements (HTTP 200).

Testing communication at the application layer proved more reliable than ICMP tests under conditions of frequent IP rotation.

2.5.4 Host Agent Logging Mechanism

A lightweight Python-based HTTP service (`host_agent.py`) was deployed on each simulated host and configured to listen on port 8080. These host agents received incoming packets and generated acknowledgment responses.

Host agent logs were used to verify:

- Successful payload delivery.
- Session identifiers for each transfer.
- Integrity verification acknowledgements.
- Bidirectional communication between nodes.

This mechanism provided additional confirmation that data transmission was successfully completed between hosts.

2.5.5 OpenFlow and NAT Flow Inspection

The OpenFlow flow tables installed within Open vSwitch (OVS) were inspected to verify the correct implementation of NAT translation rules. This analysis ensured that private IP addresses were properly translated into public MTD addresses and that packet forwarding behavior matched the expected controller logic.

Flow consistency was also verified after each IP hopping event to confirm that outdated flow entries were removed and new entries were installed correctly.

2.5.6 Connectivity and Route Verification

ICMP-based tests were periodically performed to verify route availability during IP rotations. These tests allowed the controller to confirm whether dynamically assigned public IP addresses remained reachable after configuration changes.

2.5.7 Stability Evaluation Under Dynamic IP Hopping

To evaluate the impact of MTD on network stability, several IP hopping intervals were tested using a zone-based rotation strategy:

- **Low Risk Zone**: 80 seconds
- **Medium Risk Zone**: 100 seconds
- **High Risk Zone**: 120 seconds

These experiments helped identify the balance between maintaining strong security through frequent IP changes and preserving stable communication within the network.

2.6 Summary

This chapter presented the background concepts necessary to understand the proposed system. The discussion began by examining the security limitations of traditional static hospital LAN environments and the risks associated with predictable network structures. The concept of Moving Target Defense was then introduced as a proactive strategy for reducing attacker reconnaissance capabilities.

Finally, the evaluation methodology and simulation tools used in this project were described, including the Mininet network emulator, the Ryu SDN controller, and custom host agents used for communication verification. These components formed the foundation for implementing and testing the proposed MTD-HealthNet framework, which will be described in detail in the following chapters.

Chapter 3

Literature Survey

In recent years, several studies have explored the application of Moving Target Defense (MTD) techniques to enhance network security. Traditional security systems rely on static configurations that attackers can easily analyze and exploit. In contrast, MTD dynamically changes network parameters like IP addresses, ports, and configurations to create uncertainty and limit reconnaissance success. Researchers have proposed various implementations of MTD in IoT, cloud, and software-defined networks. However, limited work has focused on its application in hospital LANs, where real-time communication and device reliability are critical.

Table 3.1 summarizes a few notable works related to MTD and their limitations:

Table 3.1: Comparison of Moving Target Defense Techniques

Technique	**Security Effectiveness**	**Overhead / Complexity**	**Typical Deployment Issues**
IP Hopping	High effectiveness against reconnaissance and worm propagation	Requires coordinated address management and client awareness	Can challenge session continuity for certain legacy applications [1] [2]
Port Hopping	High effectiveness against port-based attacks	Requires synchronization between client and server	May disrupt long-lived sessions and need protocol adaptation [4]
NAT Randomization	Provides additional protection, especially at NAT boundaries	Moderate complexity; may require customized NAT hardware or software	Needs careful timer and state management for tracking and auditability [4]

Chapter 4

Proposed Methods

In this chapter, we describe the overall design and implementation methodology used to develop the proposed MTD-HealthNet prototype. After identifying the limitations of static hospital LAN environments in the previous chapters, our objective was to build a dynamic, controller-driven network model capable of continuously modifying its attack surface while maintaining stable and secure communication between hospital devices.

Instead of relying on a purely theoretical model, the system was implemented using a practical Software Defined Networking (SDN) simulation environment. The implementation integrates dynamic IP hopping, NAT-based address translation, REST-based secure communication, and host-agent verification to replicate realistic hospital network behavior.

4.1 Architecture Overview

The proposed architecture simulates a realistic hospital Local Area Network (LAN) environment where multiple medical and administrative systems operate under strict security policies. The network is logically divided into different operational zones such as the Critical Zone, Imaging Zone, and Administrative Zone to reflect typical hospital segmentation practices.

Unlike traditional static networks, the proposed system uses a Software Defined Networking (SDN) architecture in which a centralized controller dynamically manages routing decisions, NAT mappings, and IP hopping operations.

The major architectural components are listed below:

- **Mininet-based Hospital Network:** A virtual SDN environment consisting of multiple hosts representing hospital devices such as monitoring systems, imaging servers, and staff workstations.

- **Ryu-based MTD Controller:** A centralized controller responsible for installing OpenFlow rules, performing NAT translation, enforcing security policies, and triggering dynamic IP hopping.
- **Host Agent Module:** A lightweight Python HTTP server running on each host (port 8080) used for receiving and acknowledging secure data transfers.
- **Zone Scheduler:** A policy-driven scheduling module that triggers IP rotation at predefined time intervals based on zone-specific risk levels.
- **Web Interface and Logging System:** A monitoring interface used to visualize controller logs, secure transfers, and network events for analysis and debugging.

In the implemented architecture, internal private IP addresses (10.0.0.0/24) are translated into dynamically rotating public MTD addresses (172.x.x.x range) using controller-installed NAT rules. This approach hides the internal network structure and continuously changes the externally visible attack surface.

4.2 Design Methodology

To ensure modularity and scalability, the proposed system was designed using a layered architecture rather than a monolithic structure.

4.2.1 Device Layer

The device layer represents the simulated hospital endpoints, including ICU monitoring systems, PACS imaging servers, laboratory systems, and administrative workstations.

Within the Mininet topology, each device is implemented as an independent host connected to a central OpenFlow switch. A host agent is deployed on each node to support secure communication logging and payload acknowledgment.

4.2.2 Control Layer

The control layer is implemented using a custom Ryu SDN controller that acts as the central intelligence of the MTD system. The controller dynamically manages:

- OpenFlow flow rule installation and updates
- NAT mappings between static private IP addresses and rotating public MTD IP addresses

- Zone-based access control policies
- Periodic IP hopping operations triggered by the zone scheduler

The controller logic is designed to maintain communication continuity even during IP transitions, thereby minimizing session disruption during MTD rotations.

4.2.3 Security and Monitoring Layer

To validate system behavior, multiple monitoring mechanisms were integrated into the system. These include controller logs, REST API testing, host-agent acknowledgments, and trace replay analysis. This multi-layer validation approach ensured that communication success was verified through actual data transfers rather than simple connectivity assumptions.

4.3 Network Topology and Simulation Setup

The hospital LAN environment was implemented using the Mininet network emulator. Multiple hosts (h1–h6) were connected to a central Open vSwitch that was controlled by the Ryu SDN controller. This setup allowed dynamic control of packet forwarding behavior, routing paths, and flow rule installation.

Baseline connectivity tests were initially performed using ICMP-based commands such as `pingall`. However, due to NAT translation and dynamic IP hopping, ICMP tests did not always accurately represent application-level communication success.

Therefore, system validation was performed using REST-based secure transfers and host-agent verification mechanisms.

4.4 Dynamic IP Hopping Mechanism

The Moving Target Defense engine periodically rotates public-facing IP addresses in order to invalidate attacker reconnaissance information.

The IP hopping workflow operates as follows:

1. The zone scheduler selects hosts for IP rotation based on predefined risk policies.
2. New MTD IP addresses are dynamically generated.
3. The Ryu controller updates NAT mappings and OpenFlow flow rules.

4. Host agents continue communication through translated routes without manual configuration changes.

By continuously rotating externally visible IP addresses, the system ensures that any network mapping collected by an attacker quickly becomes outdated.

4.5 NAT Randomization and Flow Rule Management

To conceal the internal network structure, NAT-based translation is performed between static private IP addresses and rotating public MTD IP addresses. The Ryu controller dynamically installs and removes OpenFlow flow entries within Open vSwitch to maintain correct packet forwarding behavior during IP transitions.

Flow tables were inspected to verify:

- Correct rewriting of source and destination IP addresses
- Removal of outdated flow entries after IP hopping
- Consistent routing paths during NAT updates

These checks ensured that the network remained synchronized during dynamic reconfiguration.

4.6 Secure Transfer and REST API Validation

To simulate realistic hospital data exchanges, a REST API endpoint (`/sim/secure_transfer`) was implemented. Secure communication between hosts was validated using repeated `cURL` requests.

The API-based testing verified:

- Policy-compliant source and destination host selection
- Successful payload transmission and acknowledgment
- HTTP response verification (HTTP 200 for successful transfers)
- Controller trace logs for routing and policy decisions

This approach provided reliable validation of application-level communication under dynamic IP hopping conditions.

4.7 Host Agent Communication Verification

A lightweight Python host agent (`host_agent.py`) was deployed on each Mininet host and configured to run continuously on port 8080. These agents received incoming secure payloads and generated acknowledgment responses.

Host agent logs were used to verify:

- Successful payload delivery
- Session identifiers for each communication exchange
- Bidirectional communication consistency
- Integrity verification acknowledgments

This mechanism ensured that communication success was confirmed through actual packet delivery rather than simulation assumptions.

4.8 Stability Testing Under Zone-Based IP Rotation

To evaluate the balance between security and network stability, IP hopping intervals were configured based on zone-specific risk levels:

- Low Risk Zone – 80 seconds
- Medium Risk Zone – 100 seconds
- High Risk Zone – 120 seconds

Experimental testing showed that extremely rapid IP rotations could occasionally cause temporary synchronization delays between the controller and host agents. Therefore, the rotation intervals were optimized to maintain both security effectiveness and communication stability.

4.9 Exception Handling and Debugging

During development, several issues such as failed transfers, NAT synchronization delays, and controller runtime exceptions were encountered. These issues were analyzed using controller logs, REST API responses, and host agent outputs.

Detailed trace-based debugging allowed identification of root causes such as delayed flow installation, rapid IP rotations, and host-agent availability issues. Addressing these problems significantly improved system stability.

4.10 Tools Used

- **Mininet** – used to emulate the SDN-based hospital LAN topology and dynamic host communication.
- **Ryu SDN Controller** – used to implement Moving Target Defense logic, NAT randomization, and OpenFlow flow rule management.
- **Python** – used for implementing the controller modules, host agents, schedulers, and automation scripts.
- **cURL and REST APIs** – used to validate secure data transfer between simulated hosts.
- **CSV Logs** – used to store communication records, controller traces, and validation results.
- **Linux Environment (Kali Linux)** – used to run Mininet simulations, controller programs, and testing scripts.
- **HTML, CSS, and JavaScript** – used to develop the web-based monitoring interface and trace visualization dashboard.

4.11 Summary

This chapter presented the complete design and implementation methodology of the proposed MTD-HealthNet system. The architecture integrates a Mininet-based SDN environment, a Ryu controller, NAT randomization, REST-based secure communication, and host-agent verification mechanisms.

Unlike traditional static networks, the proposed framework continuously modifies externally visible network parameters while maintaining stable internal communication. This dynamic behavior increases attacker uncertainty and significantly reduces the effectiveness of reconnaissance and lateral movement attacks.

The next chapter presents the experimental evaluation and performance analysis of the proposed system under different dynamic IP hopping scenarios.

Chapter 5

Contribution

This chapter outlines the main contributions of our work through the design and implementation of the MTD-HealthNet prototype. Rather than presenting a purely conceptual model, our project focuses on developing a functional system that demonstrates how Moving Target Defense can be applied within a hospital network environment. The proposed system integrates Software Defined Networking (SDN), dynamic IP hopping, NAT randomization, and application-level validation to create a network that continuously changes its externally visible structure while maintaining reliable communication between hospital devices.

5.1 Innovations in the Proposed Method

One of the primary contributions of this project is the practical implementation of a Moving Target Defense mechanism within a simulated hospital Local Area Network. Traditional hospital networks often rely on static configurations where IP addresses and network structures remain unchanged. In contrast, the proposed MTD-HealthNet system dynamically modifies the network surface using IP hopping and NAT randomization, making it significantly more difficult for attackers to maintain accurate reconnaissance data.

A key innovation in our work is the development of a custom controller based on the Ryu SDN framework. This controller manages OpenFlow rules, performs NAT translations, and enforces zone-based security policies. By dynamically updating these parameters, the controller periodically rotates the externally visible IP addresses of selected hosts while preserving the internal private addressing structure. This approach hides the true network layout from potential attackers.

Another important aspect of our method is the use of application-layer communica-

tion testing rather than relying solely on traditional ICMP-based connectivity checks. We implemented a dedicated REST API endpoint (`/sim/secure_transfer`) to simulate secure communication between hosts. This allowed us to evaluate real data exchange even when IP addresses were continuously changing.

To further strengthen the validation process, lightweight host agents were deployed on each virtual node. These agents listen on port 8080 and generate acknowledgment logs whenever a secure transfer is received. By analyzing these logs, we were able to verify that communication between hosts was successfully completed under dynamic network conditions.

5.2 Practical SDN-Based Hospital Network Simulation

Another significant contribution of this project is the development of a realistic hospital network simulation environment. Using Mininet and Open vSwitch, we constructed a virtual LAN consisting of multiple hosts representing hospital systems such as medical devices, servers, and administrative workstations.

These hosts were connected through a centralized SDN controller, which dynamically installed and updated OpenFlow rules to manage packet forwarding. Instead of exposing internal private IP addresses directly, NAT translation was applied to map private addresses (10.0.0.0/24) to dynamically changing public MTD addresses in the 172.x.x.x range. This design hides the internal network structure and continuously changes the externally visible network layout.

The simulation-based environment allowed us to evaluate the proposed system in a controlled and reproducible manner without requiring access to real hospital infrastructure.

5.3 Performance and Security Observations

During repeated testing, the system demonstrated that dynamic IP changes did not significantly disrupt communication between hosts. The controller successfully updated flow rules and NAT mappings while maintaining stable communication channels.

Key observations from the experiments include:

- **Communication Stability:** Host-to-host communication through REST-based transfers remained functional even after multiple IP rotations.

- **Minimal Disruption During IP Hopping:** Short delays were occasionally observed during IP updates, but the network quickly stabilized without manual intervention.
- **Controller Adaptability:** The Ryu controller efficiently updated flow rules and NAT mappings without introducing major routing inconsistencies.
- **Reduced Reconnaissance Effectiveness:** Continuous IP rotation made it difficult to maintain a consistent network map, thereby limiting the usefulness of attacker reconnaissance.

These results indicate that dynamic network reconfiguration can enhance security without significantly degrading communication performance.

5.4 Security Impact on Reconnaissance and Lateral Movement

From a defensive perspective, the proposed architecture reduces the effectiveness of common attack strategies. In static networks, attackers often rely on reconnaissance to map the network and identify valuable targets. Once the network structure is known, lateral movement becomes easier.

In the MTD-HealthNet system, however, dynamic IP reassignment continuously modifies the network environment. This means that reconnaissance data collected at one moment quickly becomes outdated. As a result, attackers must repeatedly restart scanning activities, increasing the time and effort required to perform a successful attack.

Additionally, NAT randomization and controller-driven policy enforcement introduce further layers of abstraction between external attackers and internal devices. This additional complexity helps protect sensitive systems from direct exposure.

5.5 Experimental Observations

During testing, we also observed some interesting system behaviors. When IP hopping occurred too frequently, brief synchronization delays between the controller and host agents were occasionally observed. These delays sometimes caused short interruptions in secure transfers. However, the system automatically stabilized once the updated flow rules and NAT mappings were applied.

Another observation was that application-layer validation proved more reliable than traditional ICMP testing in this dynamic environment. Because NAT translation and IP hopping can affect ICMP responses, `pingall` tests in Mininet sometimes indicated failures even though actual data transfers between hosts were successful.

We also found that implementing zone-based IP rotation provided a better balance between security and stability compared to very rapid global IP hopping. This approach allows high-risk zones to rotate IP addresses more frequently while maintaining stable communication in lower-risk areas.

5.6 Overall Contribution of the Project

The main contribution of this project is the development of a functional SDN-based hospital network prototype that integrates multiple security mechanisms within a single framework. The system combines dynamic IP hopping, NAT randomization, application-layer secure communication, and host-agent verification to create a continuously changing network environment.

Unlike many existing MTD studies that remain purely conceptual, our implementation demonstrates that Moving Target Defense can be practically deployed in a simulated healthcare network while maintaining stable communication between devices. The results indicate that MTD can significantly reduce the effectiveness of reconnaissance and lateral movement attacks while preserving operational reliability.

This prototype provides a foundation for further research on dynamic security architectures for healthcare networks and other critical infrastructures.

Chapter 6

Results and Analysis

6.1 Performance Evaluation

To examine the effectiveness of the proposed MTD-HealthNet system, a series of experiments were conducted using a Mininet-based Software Defined Networking (SDN) environment. The experimental setup consisted of several virtual hosts representing hospital devices, an Open vSwitch responsible for packet forwarding, and a Ryu-based controller implementing the Moving Target Defense logic.

The controller dynamically updated IP mappings at predefined time intervals depending on the zone classification of each host. During these rotations, the controller also maintained the necessary NAT translations and OpenFlow flow rules to ensure that communication between hosts remained uninterrupted.

Instead of evaluating the network solely through ICMP connectivity tests, both connectivity checks and REST-based secure transfers were performed between hosts. These application-level transfers provided a more realistic representation of communication between hospital systems under conditions where IP addresses were continuously changing.

Throughout the testing process, background network activity was generated using repeated ping requests and HTTP-based secure transfers. These operations allowed us to observe the behavior of the network during repeated IP hopping cycles.

Experimental observations showed that dynamic IP reassignment introduced only minor performance variations. Small delays were occasionally observed during the exact moment when IP mappings were updated, but communication quickly stabilized once the controller synchronized the new flow rules and NAT entries.

The following general observations were recorded:

- Packet delivery remained stable during most IP rotation cycles because the controller updated flow rules before communication was interrupted.

- A small increase in round-trip time (RTT) was observed during IP reassignment events due to temporary controller synchronization.
- Continuous IP rotation reduced the effectiveness of network reconnaissance attempts because host addresses were constantly changing.

6.1.1 Configured IP Hopping Intervals

During the final system configuration, a zone-based IP hopping strategy was adopted. Initial experiments showed that extremely frequent IP changes could occasionally introduce short synchronization delays between the controller, NAT mappings, and host agents.

To balance security with communication stability, different IP rotation intervals were assigned according to the risk level of each hospital zone.

Table 6.1: Zone-Based IP Hopping Configuration in MTD-HealthNet

Hospital Zone	**Risk Level**	**IP Hopping Interval (seconds)**
Administrative Zone	Low	80
Imaging Zone	Medium	100
Critical Zone	High	120

Repeated experiments indicated that longer intervals in high-risk zones improved controller synchronization while still maintaining the security benefits of Moving Target Defense.

Table 6.2: Comparison of Network Behavior Before and After MTD Deployment

Parameter	**Static Network**	**MTD-Enabled Network**
Packet Delivery Ratio	Nearly 100%	Slight variation during IP rotation
Average RTT	Stable and constant	Small increase during hopping events
Reconnaissance Success Rate	High due to static hosts	Reduced due to changing IP mappings
Connection Stability	Stable	Stable after controller updates

6.2 Impact of Dynamic IP Hopping

One of the main goals of this project was to evaluate how continuous IP randomization affects communication reliability within a hospital LAN environment.

Experimental results showed that host-to-host communication remained functional even after multiple IP rotations. This was possible because the SDN controller dynamically updated OpenFlow rules and NAT mappings whenever an IP change occurred.

Although standard `pingall` tests in Mininet occasionally showed inconsistent ICMP responses during rapid IP changes, REST-based secure transfer tests and host agent logs confirmed that actual data transmission between hosts remained successful.

Each host ran a lightweight agent on port 8080 that generated acknowledgment logs whenever a secure transfer was received. These logs provided clear confirmation that communication was preserved even when the external IP addresses were modified.

6.3 Strengths and Limitations

Strengths

- The system provides proactive defense by continuously modifying the externally visible network structure.
- Controller-managed flow rule updates maintain communication stability during IP hopping.
- The evaluation methodology uses realistic application-layer testing instead of relying solely on network-layer connectivity checks.
- Zone-based IP rotation helps maintain a balance between security strength and network stability.
- The SDN-based architecture allows the approach to scale to larger hospital network environments.

Limitations

- Very rapid IP rotation may temporarily delay synchronization between the controller and host agents.
- Some legacy medical systems that rely on fixed IP addresses may require additional configuration.
- Small variations in latency can occur during IP reassignment cycles.
- Continuous operation of the SDN controller is necessary to maintain NAT mappings and policy enforcement.

6.4 Security Analysis

From a security standpoint, the proposed architecture significantly reduces the effectiveness of reconnaissance and network mapping attacks. In a traditional static network, attackers can gradually identify active hosts and services and then use this information to perform lateral movement.

In the MTD-HealthNet environment, however, IP addresses change periodically. As a result, any reconnaissance data collected by an attacker becomes outdated within a short time period.

Experimental observations during simulated scanning activities showed that host discovery results became inconsistent over time because the mapped IP addresses changed frequently. Additionally, NAT randomization concealed the internal network structure, making it difficult for attackers to determine the true location of critical devices.

Controller logs, REST API responses, and host agent acknowledgments were analyzed during testing. These records confirmed that legitimate communications continued successfully while unauthorized or inconsistent connections were disrupted during IP transitions.

Overall, the experimental results demonstrate that the MTD-HealthNet system improves resilience against reconnaissance and lateral movement attacks while maintaining acceptable communication performance and service stability within the simulated hospital network environment.

Chapter 7

Conclusion

This project investigated the use of Moving Target Defense (MTD) as a proactive approach to improve the security of hospital network environments. Traditional hospital LANs typically rely on static configurations, which makes them vulnerable to reconnaissance and network mapping attacks. To address this issue, we developed an SDN-based prototype called **MTD-HealthNet**, where network parameters are dynamically modified to reduce predictability.

The system was implemented in a simulated environment using Mininet, Open vSwitch, and a Ryu-based controller. Through dynamic IP hopping and NAT-based address translation, the externally visible network structure was continuously changed while internal communication between hosts remained stable. Secure transfers and host-agent verification were used to confirm that communication continued successfully during IP reconfiguration.

Experimental observations showed that periodic IP rotation reduced the effectiveness of reconnaissance attempts while maintaining acceptable communication stability. The controller successfully synchronized flow rules and NAT mappings during each IP transition, resulting in only minor and temporary performance variations.

Overall, the project demonstrates that Moving Target Defense can enhance the resilience of hospital networks by making the network environment less predictable to attackers while preserving reliable communication between devices.

Future work during the later stages of this project primarily focused on debugging the controller code, stabilizing the system during dynamic IP hopping operations, and documenting the implementation and experimental results in this report.

Chapter 8

Future Work

The current implementation of MTD-HealthNet demonstrates a working proof-of-concept of a Moving Target Defense mechanism for hospital networks using SDN and dynamic IP hopping. Although the prototype successfully validated the feasibility of the approach, several aspects require further investigation and refinement.

Future work will primarily focus on improving the stability of the system through additional debugging and optimization of the controller logic. During experimentation, some synchronization delays were observed during rapid IP transitions, and further refinement of the code can help improve reliability during dynamic reconfiguration.

Another important direction is performing a more detailed performance analysis of the system. This includes evaluating throughput, latency variations, and packet delivery behavior during IP hopping cycles. In particular, future experiments should quantify packet loss that may occur during the transition between IP mappings and analyze how SDN flow rule updates influence network performance.

Controller scalability is also an important aspect that requires deeper study. Measuring the CPU and memory overhead of the Ryu controller while increasing the number of hosts or the frequency of IP rotations would provide valuable insights into the system's scalability.

In addition to IP hopping, future work can explore the implementation of dynamic port hopping. Early attempts revealed challenges related to maintaining transport-layer session states. Further research could investigate methods for transparent port rewriting using the SDN controller so that applications remain unaffected by port changes.

A comparative security study can also be conducted between static networks and MTD-enabled networks. Metrics such as scanning success rate or mean time required for an attacker to map the network can be used to quantitatively evaluate the reduction in attack surface provided by the proposed system.

Finally, the collected experimental results and system improvements can be extended into a full research publication. With more detailed performance measurements and comparative security analysis, the proposed MTD-HealthNet framework can serve as a foundation for future academic work in dynamic network defense mechanisms.

References

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[5] C.-W. Huang, I.-H. Liu, J.-S. Li, C.-C. Wu, C.-F. Li, and C.-G. Liu, "A Legacy Infrastructure-based Mechanism for Moving Target Defense," *2021 IEEE 3rd Eurasia Conference on Biomedical Engineering, Healthcare and Sustainability (ECBIOS)*, IEEE, 2021. Available: http://doi.org/10.1109/ecbios51820.2021.9510261.

Appendix A

Source Code

This appendix provides access to the main implementation files used in the project. Due to the large size of the source code files, they are maintained in a public GitHub repository.

A.1 MTD Controller (Ryu SDN)

The implementation of the Moving Target Defense controller developed using the Ryu SDN framework is available at the following repository:

https://github.com/kishansaiyallanuru/MTD/blob/main/mtd_controller.py

A.2 Mininet Network Topology

The Mininet topology used to simulate the hospital network environment is available at:

https://github.com/kishansaiyallanuru/MTD/blob/main/mininet_topo.py

A.3 Policy Debugging Tool

The policy debugging tool used for validating access control rules is available at:

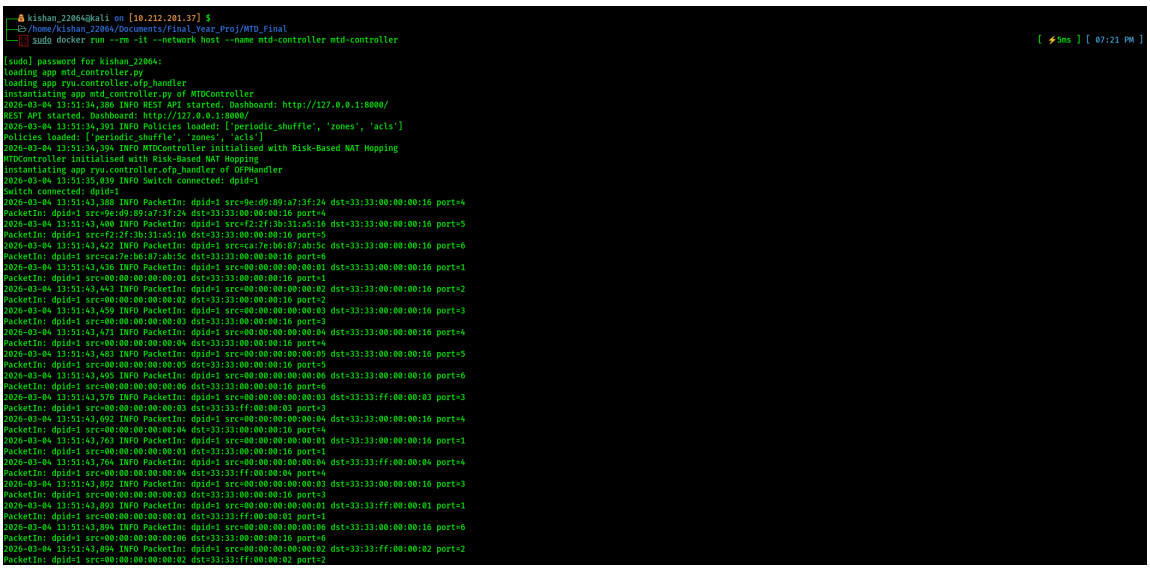
https://github.com/kishansaiyallanuru/MTD/blob/main/debug_policy.py

Appendix B

Screenshots

This appendix presents selected screenshots demonstrating the execution and functionality of the proposed MTD-HealthNet system.

B.1 MTD Controller Initialization

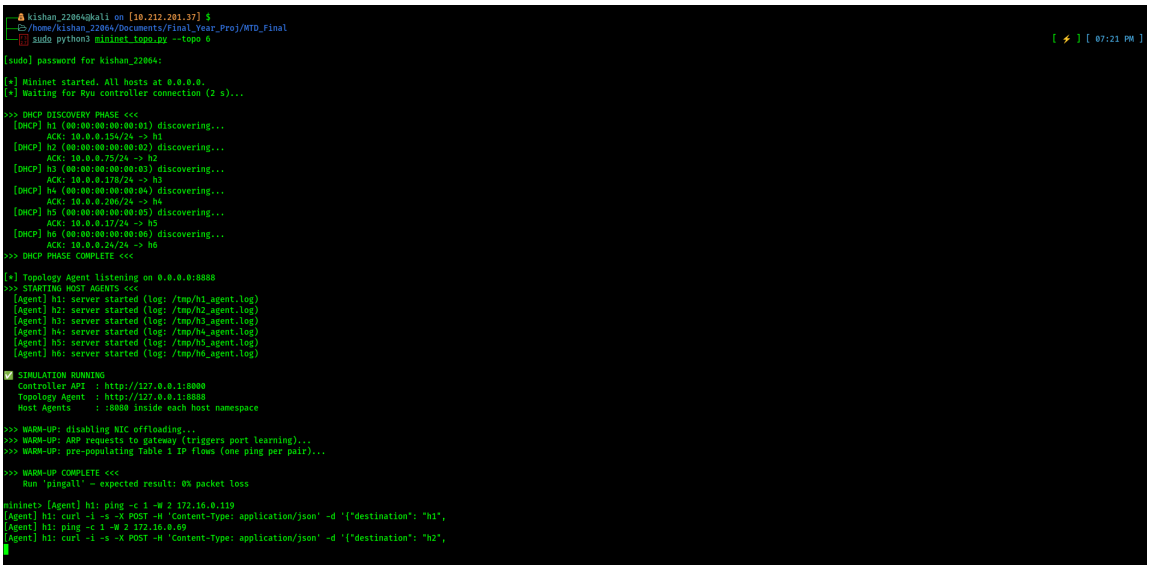


```
┌──(kishan_22064㉿kali) on [10.212.201.37] $
│ /home/kishan_22064/Documents/Final_Year_Proj/MTD_Final
└─$ sudo docker run --rm -it --network host --name mtd-controller mtd-controller                    [ ⚡5ms ] [ 07:21 PM ]

[sudo] password for kishan_22064:
loading app mtd_controller.py
loading app ryu.controller.ofp_handler
instantiating app mtd_controller.py of MTDController
2026-03-04 13:51:34,386 INFO REST API started, Dashboard: http://127.0.0.1:8000/
REST API started, Dashboard: http://127.0.0.1:8000/
2026-03-04 13:51:34,391 INFO Policies loaded: ['periodic_shuffle', 'zones', 'acls']
Policies loaded: ['periodic_shuffle', 'zones', 'acls']
2026-03-04 13:51:34,394 INFO MTDController initialised with Risk-Based NAT Hopping
MTDController initialised with Risk-Based NAT Hopping
instantiating app ryu.controller.ofp_handler of OFPHandler
2026-03-04 13:51:35,039 INFO Switch connected: dpid=1
Switch connected: dpid=1
2026-03-04 13:51:43,388 INFO PacketIn: dpid=1 src=9e:d9:89:a7:3f:24 dst=33:33:00:00:00:16 port=4
PacketIn: dpid=1 src=9e:d9:89:a7:3f:24 dst=33:33:00:00:00:16 port=4
2026-03-04 13:51:43,400 INFO PacketIn: dpid=1 src=f2:2f:3b:31:a5:16 dst=33:33:00:00:00:16 port=5
PacketIn: dpid=1 src=f2:2f:3b:31:a5:16 dst=33:33:00:00:00:16 port=5
2026-03-04 13:51:43,422 INFO PacketIn: dpid=1 src=ca:7e:b6:87:ab:5c dst=33:33:00:00:00:16 port=6
PacketIn: dpid=1 src=ca:7e:b6:87:ab:5c dst=33:33:00:00:00:16 port=6
2026-03-04 13:51:43,436 INFO PacketIn: dpid=1 src=00:00:00:00:00:01 dst=33:33:00:00:00:16 port=1
PacketIn: dpid=1 src=00:00:00:00:00:01 dst=33:33:00:00:00:16 port=1
2026-03-04 13:51:43,443 INFO PacketIn: dpid=1 src=00:00:00:00:00:02 dst=33:33:00:00:00:16 port=2
PacketIn: dpid=1 src=00:00:00:00:00:02 dst=33:33:00:00:00:16 port=2
2026-03-04 13:51:43,459 INFO PacketIn: dpid=1 src=00:00:00:00:00:03 dst=33:33:00:00:00:16 port=3
PacketIn: dpid=1 src=00:00:00:00:00:03 dst=33:33:00:00:00:16 port=3
2026-03-04 13:51:43,471 INFO PacketIn: dpid=1 src=00:00:00:00:00:04 dst=33:33:00:00:00:16 port=4
PacketIn: dpid=1 src=00:00:00:00:00:04 dst=33:33:00:00:00:16 port=4
2026-03-04 13:51:43,483 INFO PacketIn: dpid=1 src=00:00:00:00:00:05 dst=33:33:00:00:00:16 port=5
PacketIn: dpid=1 src=00:00:00:00:00:05 dst=33:33:00:00:00:16 port=5
2026-03-04 13:51:43,495 INFO PacketIn: dpid=1 src=00:00:00:00:00:06 dst=33:33:00:00:00:16 port=6
PacketIn: dpid=1 src=00:00:00:00:00:06 dst=33:33:00:00:00:16 port=6
2026-03-04 13:51:43,576 INFO PacketIn: dpid=1 src=00:00:00:00:00:03 dst=33:33:ff:00:00:03 port=3
PacketIn: dpid=1 src=00:00:00:00:00:03 dst=33:33:ff:00:00:03 port=3
2026-03-04 13:51:43,692 INFO PacketIn: dpid=1 src=00:00:00:00:00:04 dst=33:33:00:00:00:16 port=4
PacketIn: dpid=1 src=00:00:00:00:00:04 dst=33:33:00:00:00:16 port=4
2026-03-04 13:51:43,763 INFO PacketIn: dpid=1 src=00:00:00:00:00:01 dst=33:33:00:00:00:16 port=1
PacketIn: dpid=1 src=00:00:00:00:00:01 dst=33:33:00:00:00:16 port=1
2026-03-04 13:51:43,764 INFO PacketIn: dpid=1 src=00:00:00:00:00:04 dst=33:33:ff:00:00:04 port=4
PacketIn: dpid=1 src=00:00:00:00:00:04 dst=33:33:ff:00:00:04 port=4
2026-03-04 13:51:43,892 INFO PacketIn: dpid=1 src=00:00:00:00:00:03 dst=33:33:00:00:00:16 port=3
PacketIn: dpid=1 src=00:00:00:00:00:03 dst=33:33:00:00:00:16 port=3
2026-03-04 13:51:43,893 INFO PacketIn: dpid=1 src=00:00:00:00:00:01 dst=33:33:ff:00:00:01 port=1
PacketIn: dpid=1 src=00:00:00:00:00:01 dst=33:33:ff:00:00:01 port=1
2026-03-04 13:51:43,894 INFO PacketIn: dpid=1 src=00:00:00:00:00:06 dst=33:33:00:00:00:16 port=6
PacketIn: dpid=1 src=00:00:00:00:00:06 dst=33:33:00:00:00:16 port=6
2026-03-04 13:51:43,894 INFO PacketIn: dpid=1 src=00:00:00:00:00:02 dst=33:33:ff:00:00:02 port=2
PacketIn: dpid=1 src=00:00:00:00:00:02 dst=33:33:ff:00:00:02 port=2
```

Figure B.1: Startup of the MTD controller inside the Docker container. The logs show initialization of the Ryu SDN controller, loading of policies (zones, ACLs, periodic shuffle), and successful connection of the OpenFlow switch.

B.2 Hospital Network Topology Execution



```
┌──(kishan_22064㉿kali) on [10.212.201.37] $
├──╼ /home/kishan_22064/Documents/Final_Year_Proj/MTD_Final
└─╼ $ sudo python3 mininet_topo.py --topo 6                                                    [ ⚡ ] [ 07:21 PM ]

[sudo] password for kishan_22064:

[*] Mininet started. All hosts at 0.0.0.0.
[*] Waiting for Ryu controller connection (2 s)...

>>> DHCP DISCOVERY PHASE <<<
  [DHCP] h1 (00:00:00:00:00:01) discovering...
         ACK: 10.0.0.154/24 -> h1
  [DHCP] h2 (00:00:00:00:00:02) discovering...
         ACK: 10.0.0.75/24 -> h2
  [DHCP] h3 (00:00:00:00:00:03) discovering...
         ACK: 10.0.0.170/24 -> h3
  [DHCP] h4 (00:00:00:00:00:04) discovering...
         ACK: 10.0.0.206/24 -> h4
  [DHCP] h5 (00:00:00:00:00:05) discovering...
         ACK: 10.0.0.17/24 -> h5
  [DHCP] h6 (00:00:00:00:00:06) discovering...
         ACK: 10.0.0.24/24 -> h6
>>> DHCP PHASE COMPLETE <<<

[*] Topology Agent listening on 0.0.0.0:8888
>>> STARTING HOST AGENTS <<<
  [Agent] h1: server started (log: /tmp/h1_agent.log)
  [Agent] h2: server started (log: /tmp/h2_agent.log)
  [Agent] h3: server started (log: /tmp/h3_agent.log)
  [Agent] h4: server started (log: /tmp/h4_agent.log)
  [Agent] h5: server started (log: /tmp/h5_agent.log)
  [Agent] h6: server started (log: /tmp/h6_agent.log)

✅ SIMULATION RUNNING
   Controller API  : http://127.0.0.1:8000
   Topology Agent  : http://127.0.0.1:8888
   Host Agents     : :8080 inside each host namespace

>>> WARM-UP: disabling NIC offloading...
>>> WARM-UP: ARP requests to gateway (triggers port learning)...
>>> WARM-UP: pre-populating Table 1 IP flows (one ping per pair)...

>>> WARM-UP COMPLETE <<<
    Run 'pingall' — expected result: 0% packet loss

mininet> [Agent] h1: ping -c 1 -W 2 172.16.0.119
[Agent] h1: curl -i -s -X POST -H 'Content-Type: application/json' -d '{"destination": "h1",
[Agent] h1: ping -c 1 -W 2 172.16.0.69
[Agent] h1: curl -i -s -X POST -H 'Content-Type: application/json' -d '{"destination": "h2",
```

Figure B.2: Execution of the Mininet-based hospital network topology. Hosts are assigned IP addresses through DHCP, host agents are initialized, and the simulation environment is started successfully.

B.3 MTD System Monitoring Dashboard

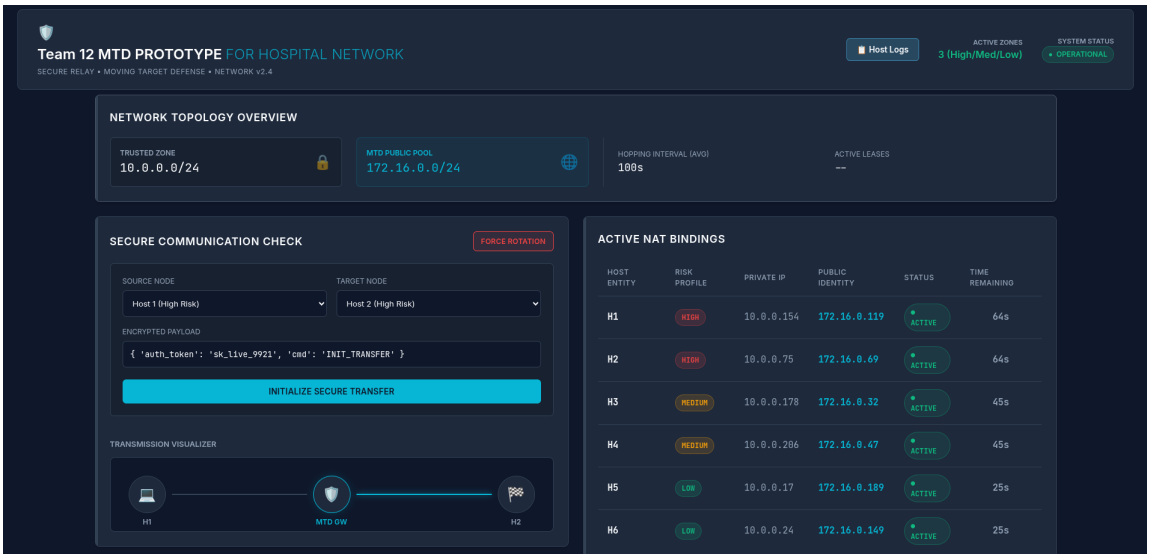


Figure B.3: Web-based monitoring interface of the MTD-HealthNet system. The dashboard shows network zones, IP hopping intervals, and current NAT bindings between private and dynamically assigned public MTD IP addresses.

B.4 Secure Communication and IP Rotation

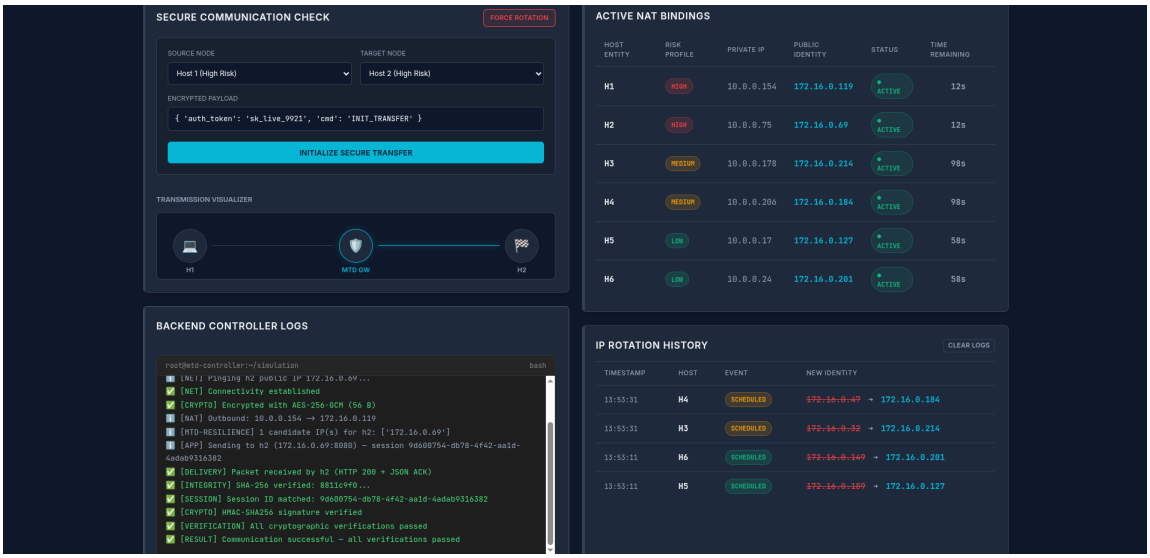


Figure B.4: Secure communication validation and IP rotation logs. The system demonstrates encrypted payload transfer, NAT translation, successful packet delivery verification, and periodic IP hopping events.

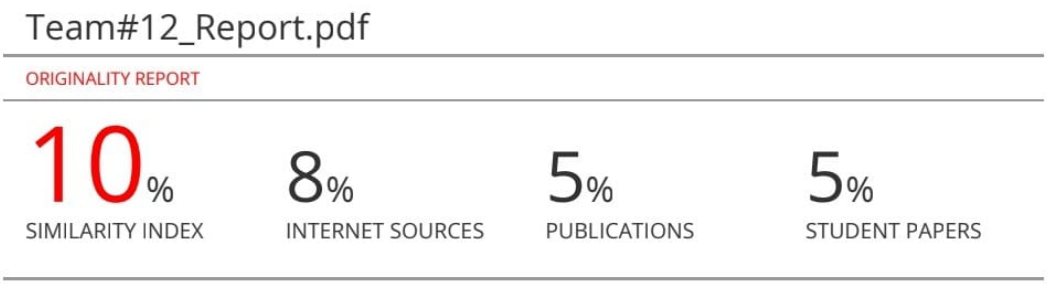
Plagiarism Report

This section includes the plagiarism reports generated for the B.Tech project document. As per academic requirements, two separate reports are provided:

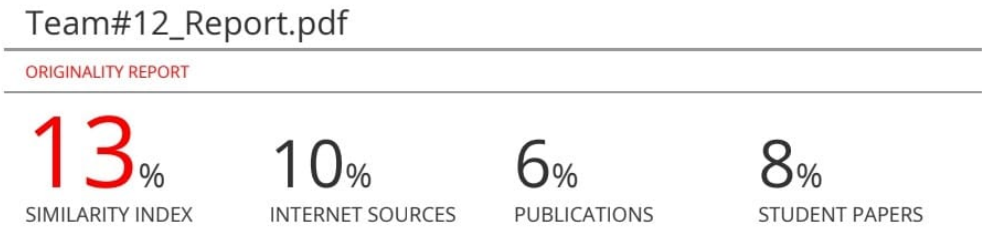
- **With Reference:** The report includes all cited and referenced materials to reflect the actual similarity index in context.
- **Without Reference:** The report excludes references and citations to assess the core originality of the submitted content.

The similarity index is within the acceptable limit prescribed by the university's academic policies, indicating that the report maintains academic integrity.

With Reference



Without Reference



Contribution to PSOs & Sustainable Development Goals

Title of the Project: MTD-HealthNet: A Moving Target Defense Model for Securing Hospital Networks
Project Members Name & Register Numbers: Burla Rushyendra Reddy (CB.EN.U4CYS22015), Karri Jeeshitha (CB.EN.U4CYS22032), Vajjula Satya Siddardha (CB.EN.U4CYS22063), Yallanuru Kishan Sai (CB.EN.U4CYS22064)
Guide Name & Designation: Mr. Akhil K J, Assistant Professor.

Contribution to PSOs:

- **PSO 1 – Cyber Security Foundations:** The project applies core cybersecurity concepts such as network defense, dynamic protection, and proactive threat mitigation. By implementing MTD in a simulated hospital environment, it reinforces the understanding of network security principles and defense mechanisms against reconnaissance and intrusion.

- **PSO 2 – Design and Implementation of Secure Systems:** Through the integration of tools like Cisco Packet Tracer and Mininet, the project showcases the ability to design, configure, and test secure network infrastructures. The implementation of a real-time IP/port randomization engine demonstrates competence in designing secure, adaptive, and scalable systems.

- **PSO 3 – Analysis and Risk Mitigation:** The work includes identifying vulnerabilities in static hospital LANs and mitigating them using MTD strategies. This aligns with analytical and evaluative skills essential for assessing cyber risks, proposing countermeasures, and validating system resilience.

The project, therefore, not only fulfills academic requirements but also enhances practical cybersecurity capabilities relevant to both industry and research applications.

Contribution to SDGs:

SDG	Contribution
3. Good Health and Well-being	Enhances cybersecurity in healthcare systems to protect patient data and ensure availability of critical systems.
9. Industry, Innovation and Infrastructure	Promotes resilient infrastructure and fosters innovation in secure digital services.
16. Peace, Justice and Strong Institutions	Supports the development of secure systems that reduce cybercrime and build trust in institutions.

SDG 3

SDG 9

SDG 16